

Rendering algorithms

Raghavendra G S & PJ Narayanan

Rendering



Rendering

Figure out which portion of the object corresponds to which pixel on the screen.



Rendering

Two broad ways

1. Rasterization
2. Ray tracing



Rendering

Two broad ways

1. Rasterization

- 1.1. Painter's algorithm
- 1.2. Z-buffer
- 1.3. A-Buffer
- 1.4. REYES

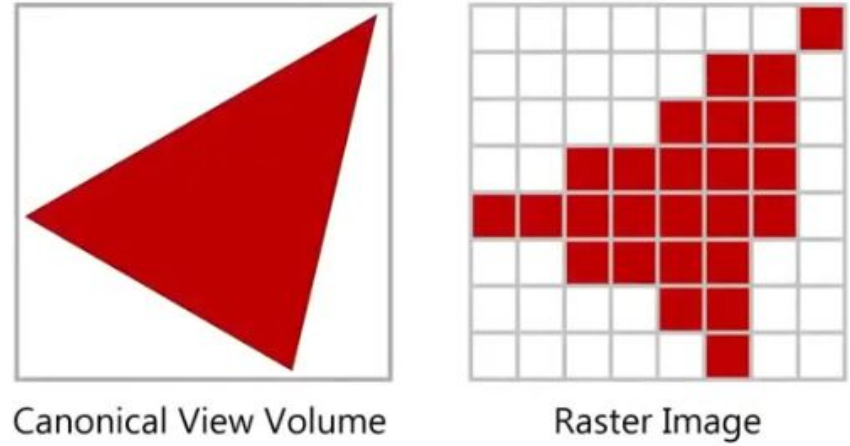
2. Ray tracing

- 2.1. Ray casting
- 2.2. Path tracing
- 2.3. ...



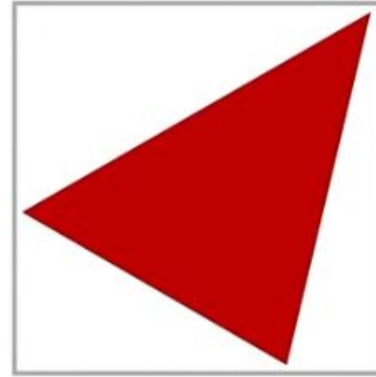
Rasterization

The process

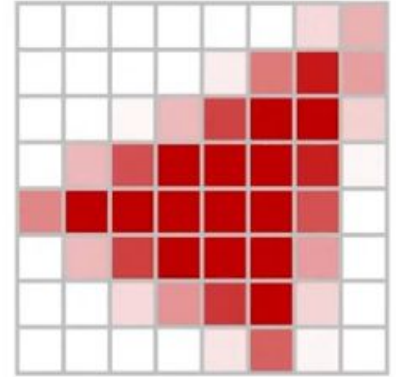


Rasterization

Artifacts?



Canonical View Volume

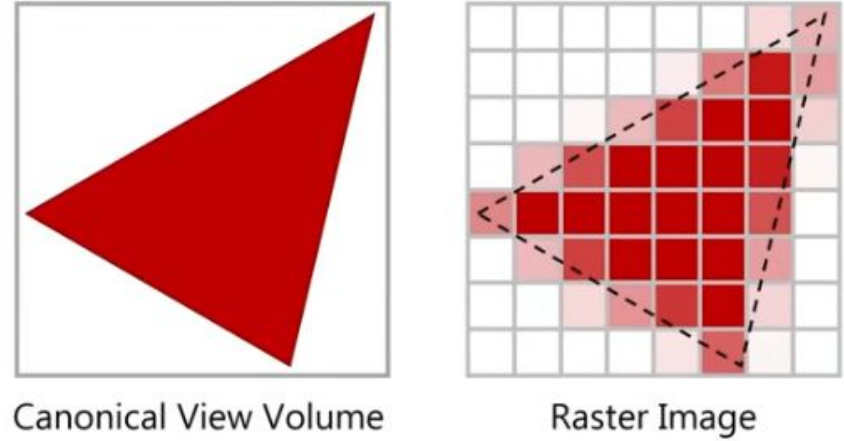


Raster Image



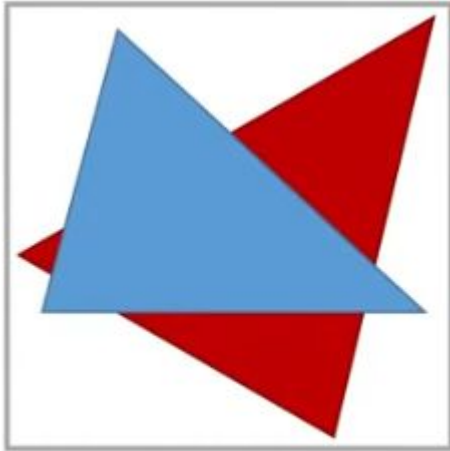
Rasterization

Antialiasing

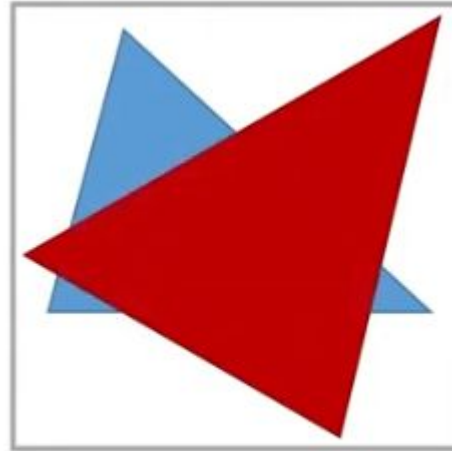


The fundamental problem

Visible surface detection or *Hidden surface removal*



Canonical View Volume



Canonical View Volume

Painter's algorithm

Draw farther objects followed by
nearer objects



Painter's algorithm

Draw farther objects followed by
nearer objects



Painter's algorithm

Draw farther objects followed by
nearer objects



Painter's algorithm

Draw farther objects followed by
nearer objects



Painter's algorithm

Draw farther objects followed by
nearer objects



Painter's algorithm

Draw farther objects followed by nearer objects



Painter's algorithm

Draw farther objects followed by nearer objects



Painter's algorithm

Draw farther objects followed by
nearer objects



Painter's algorithm

Draw farther objects followed by nearer objects



Painter's algorithm

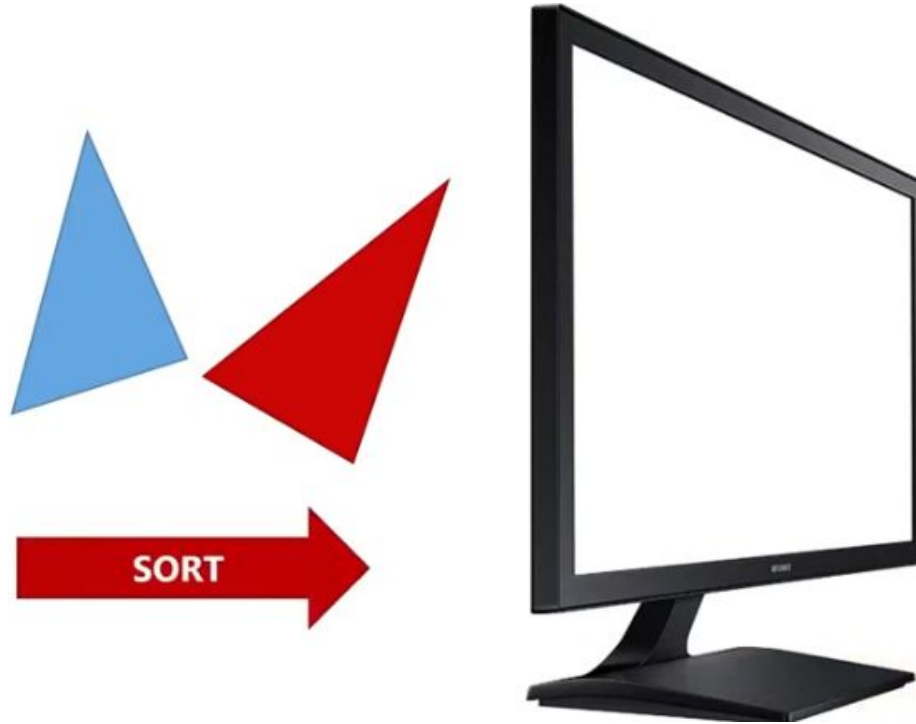
Draw farther objects followed by
nearer objects



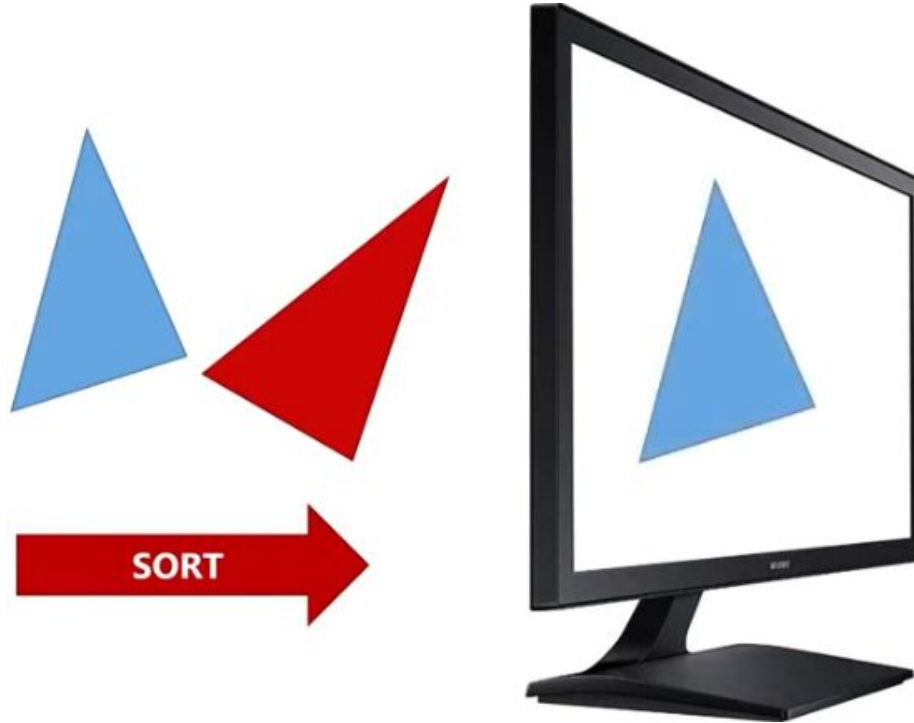
On a computer



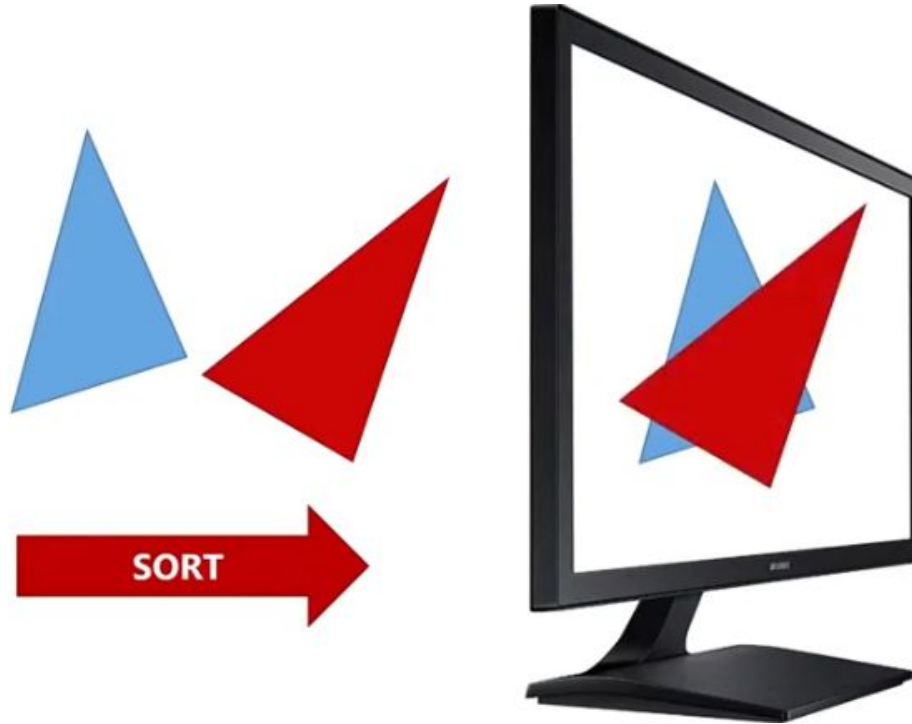
On a computer



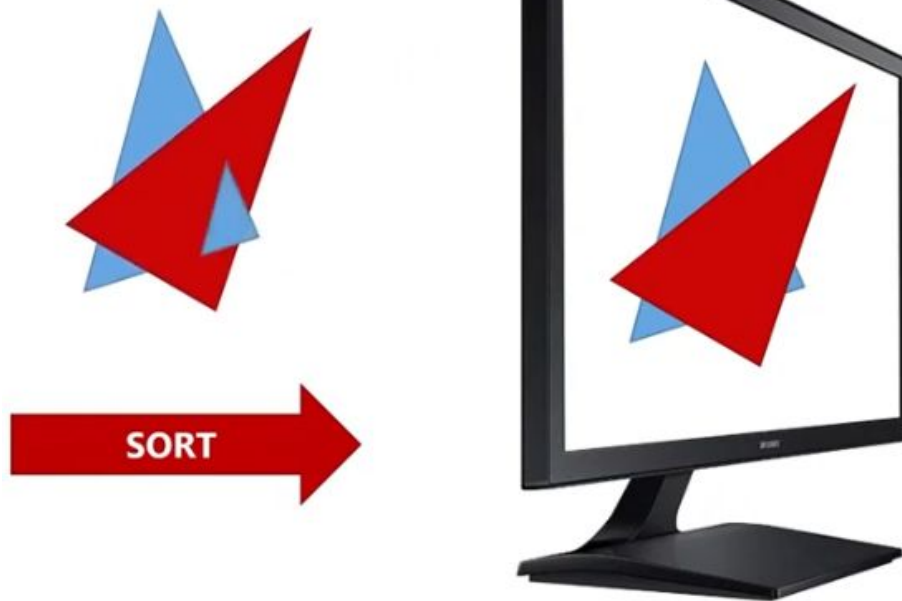
On a computer



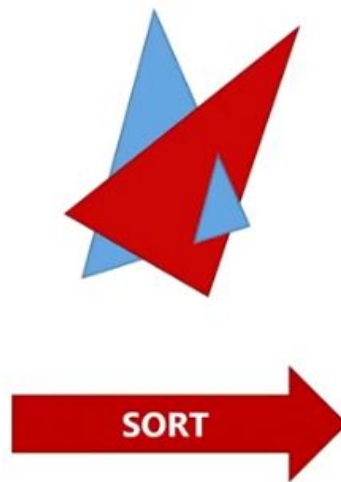
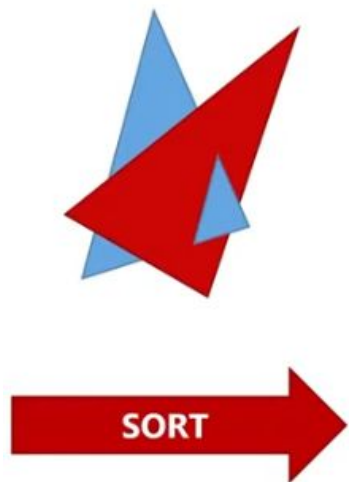
On a computer



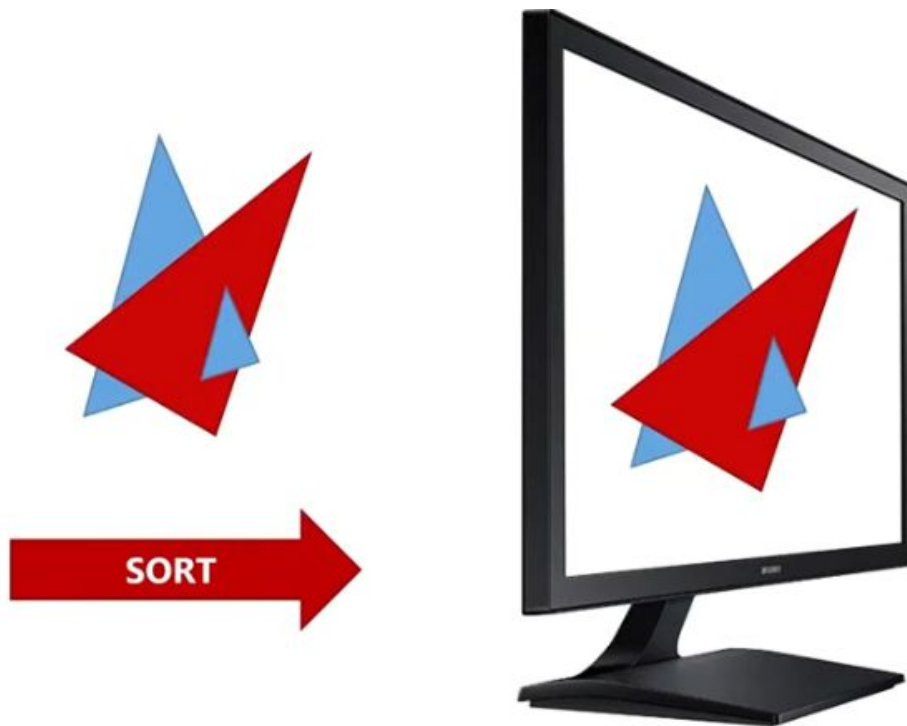
Problem/s?



Problem/s?



Problem/s?



Problem/s?

Needs sorting

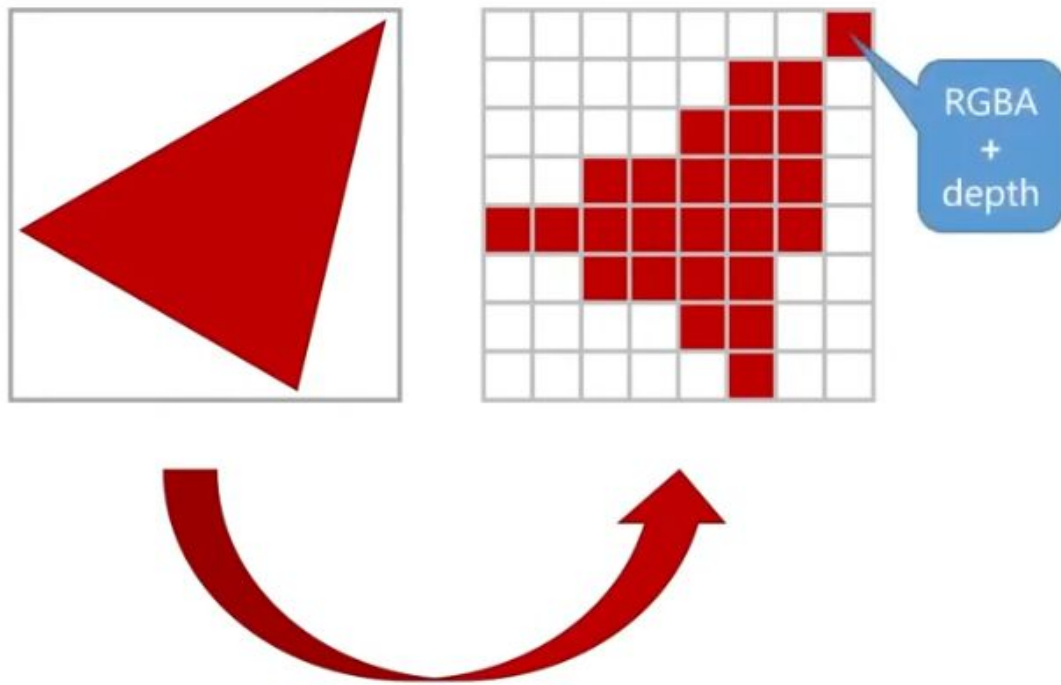
Can't handle intersecting primitives

Z-buffer

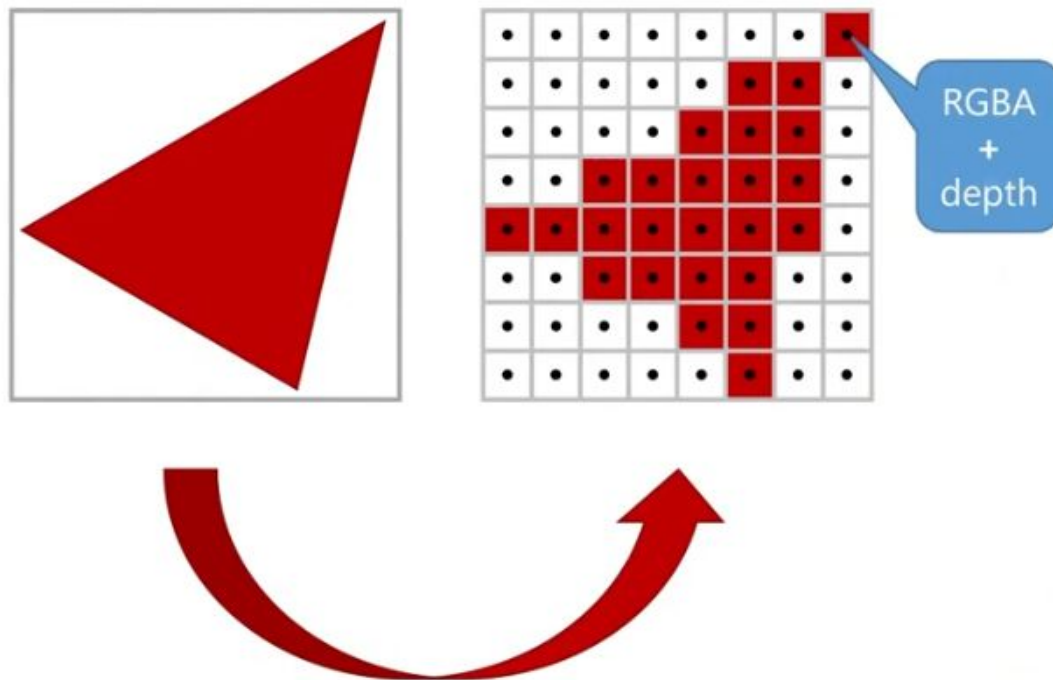
The default algorithm used in almost all the graphic pipelines

GPUs are designed to run Z-buffer

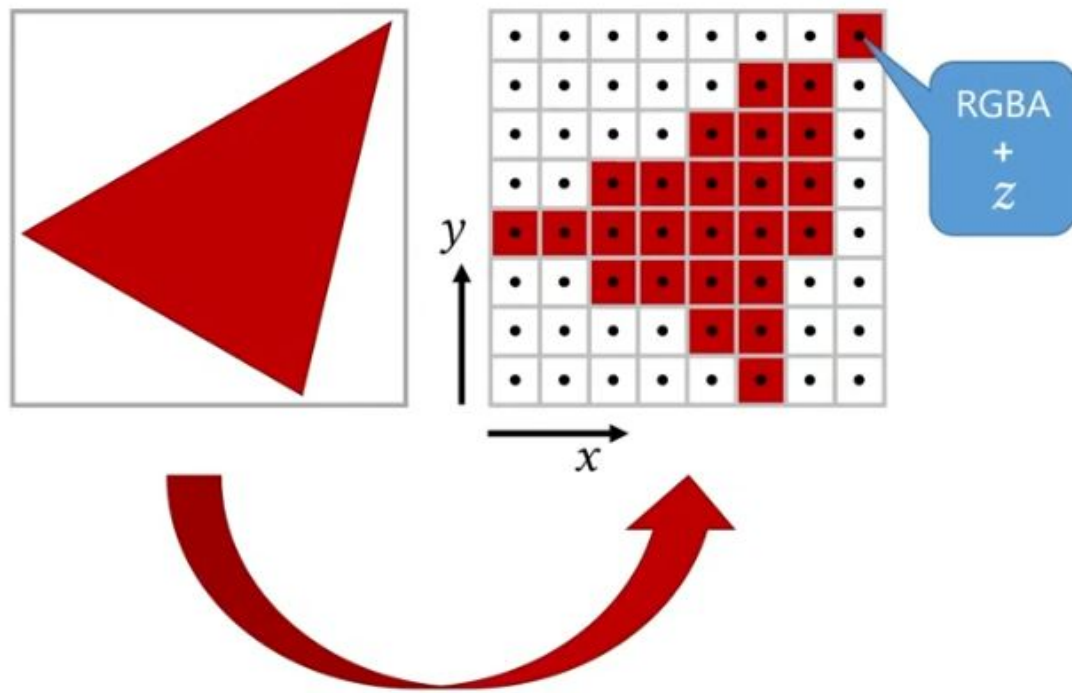
Z-buffer



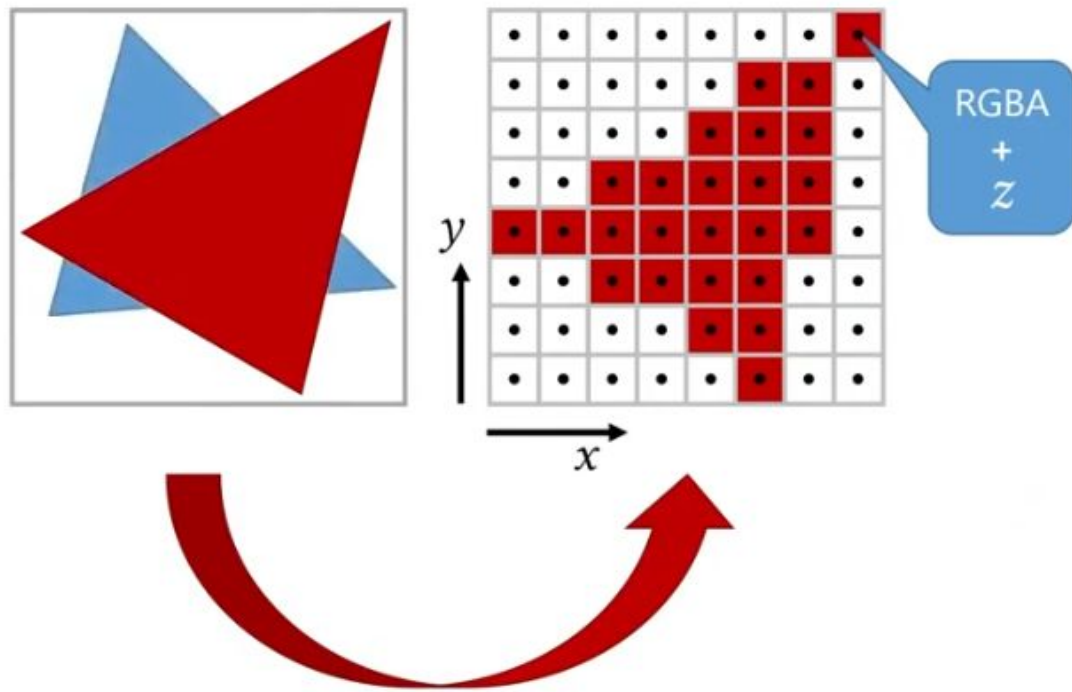
Z-buffer



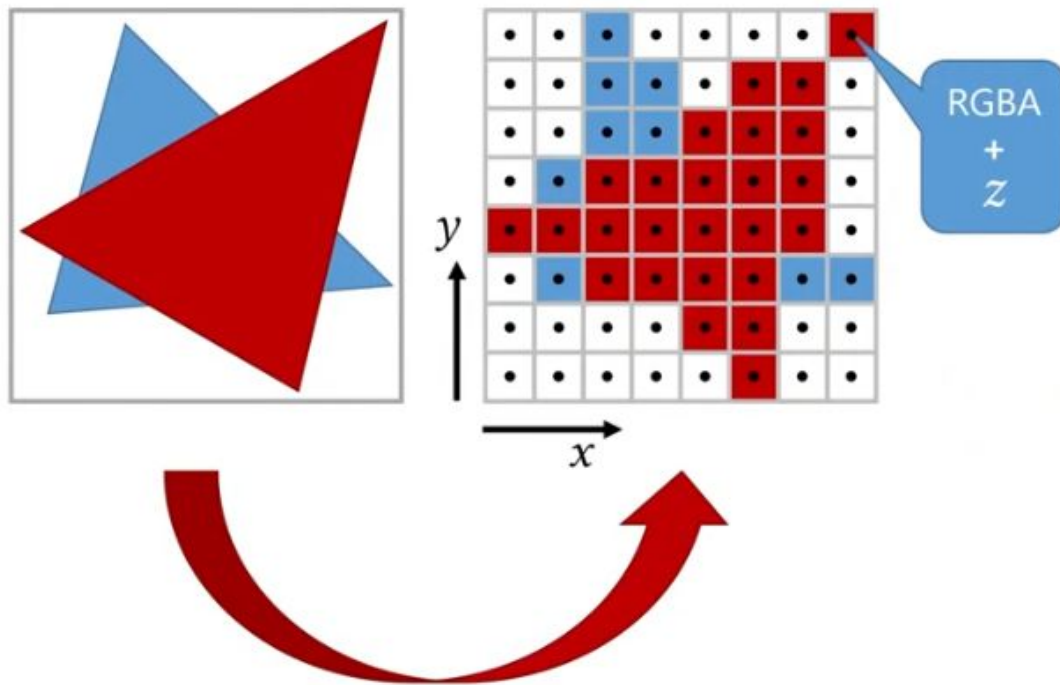
Z-buffer



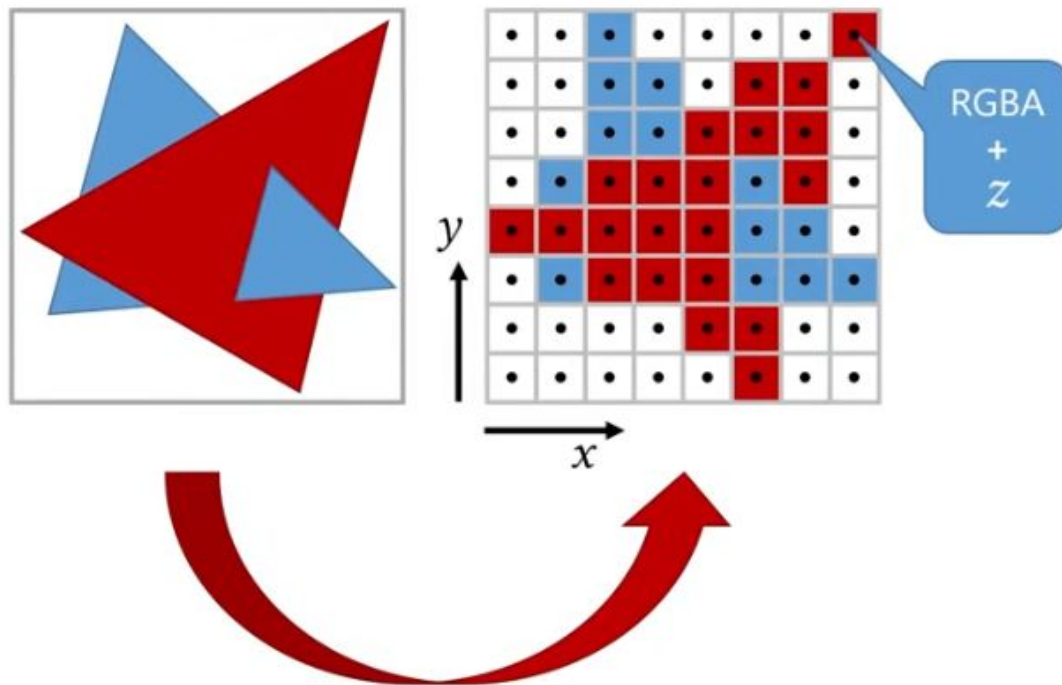
Z-buffer



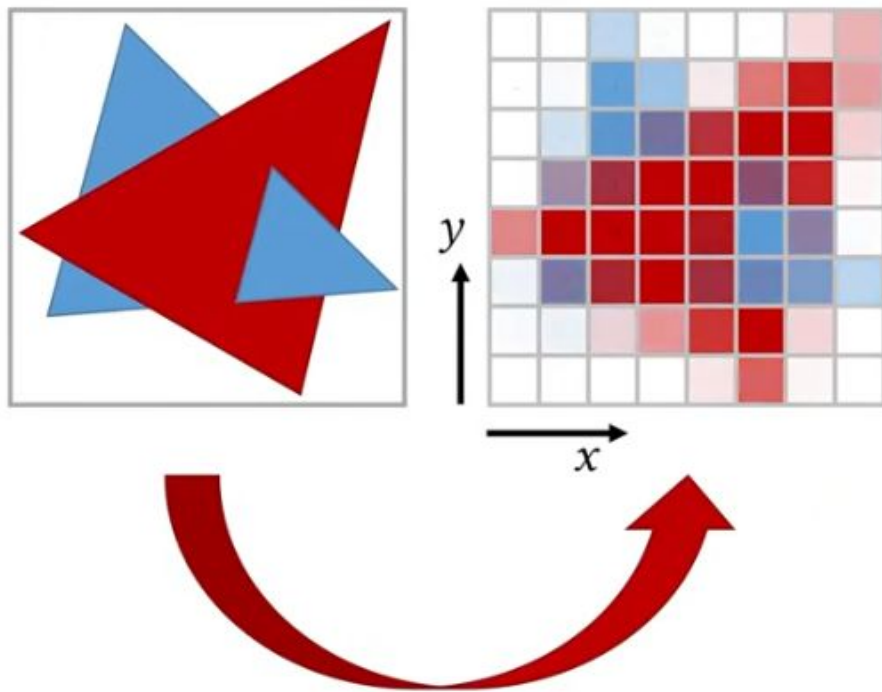
Z-buffer



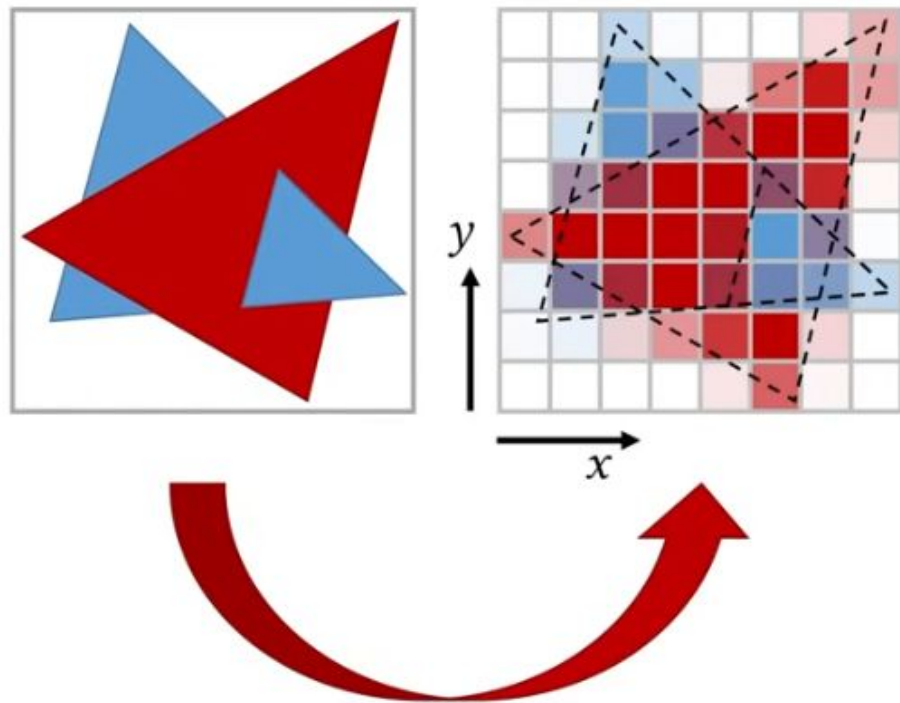
Z-buffer



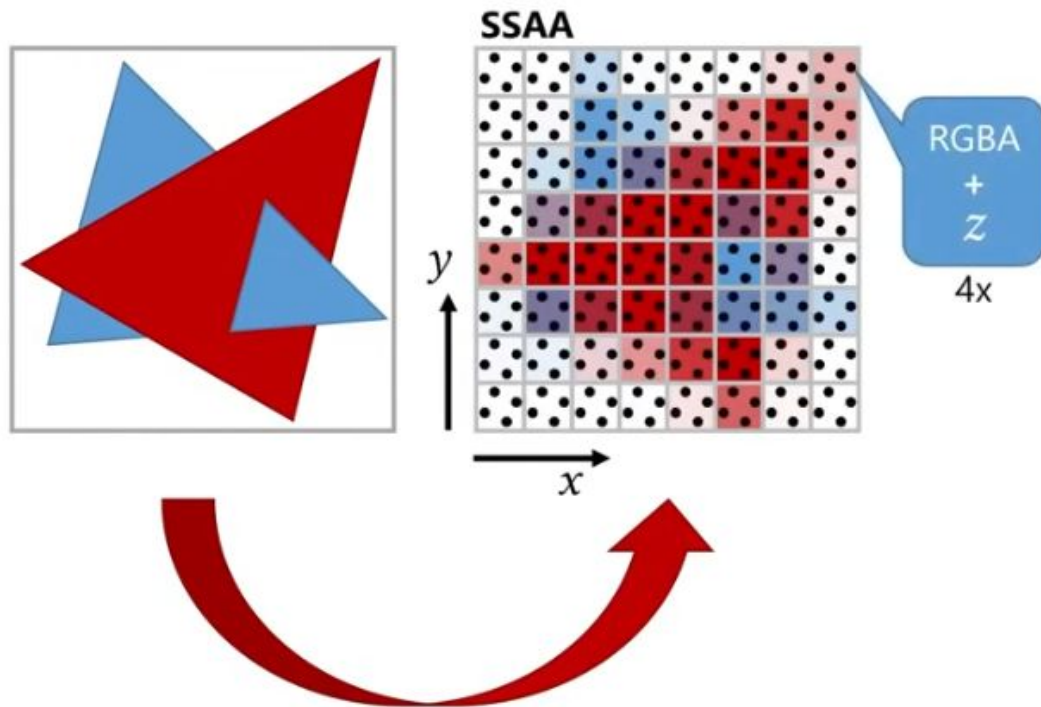
Z-buffer



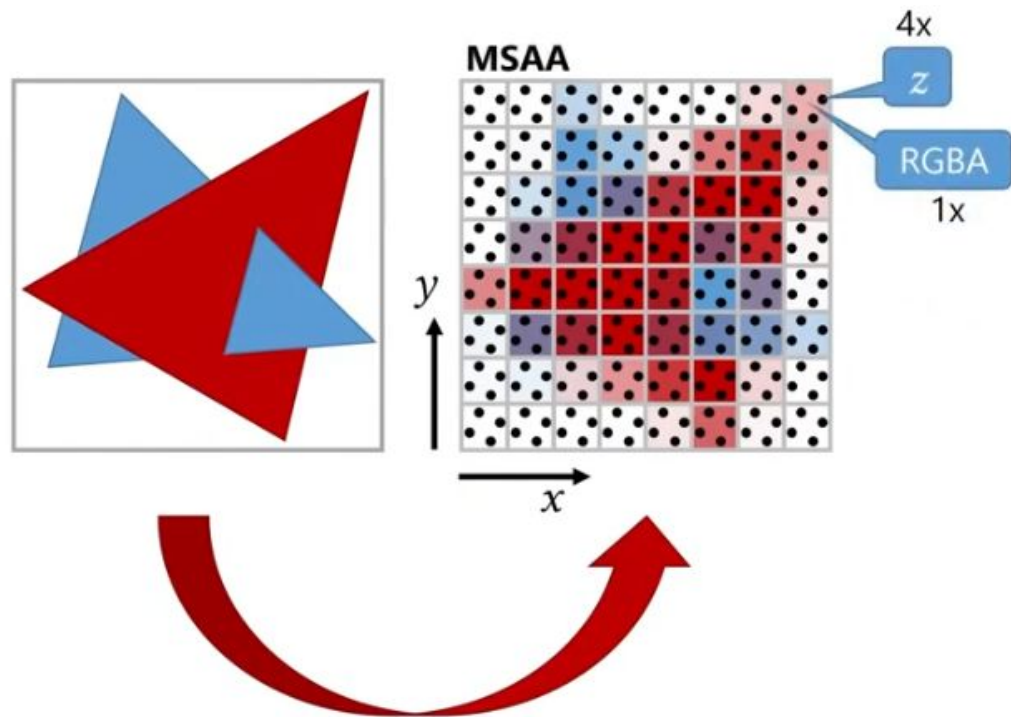
Z-buffer



Z-buffer antialiasing

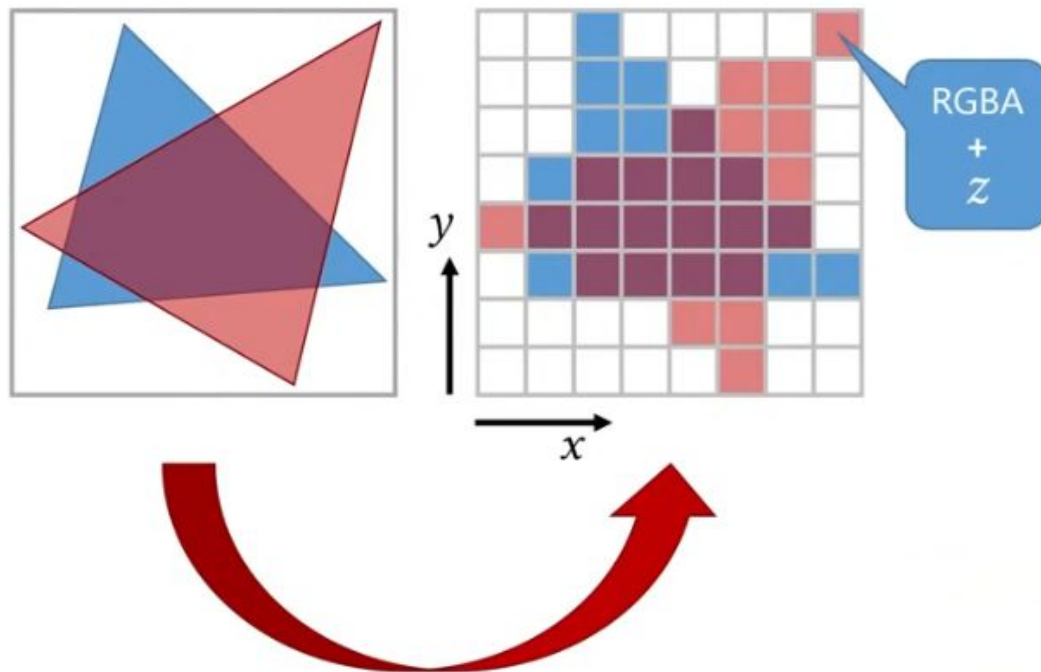


Z-buffer antialiasing



Problem/s?

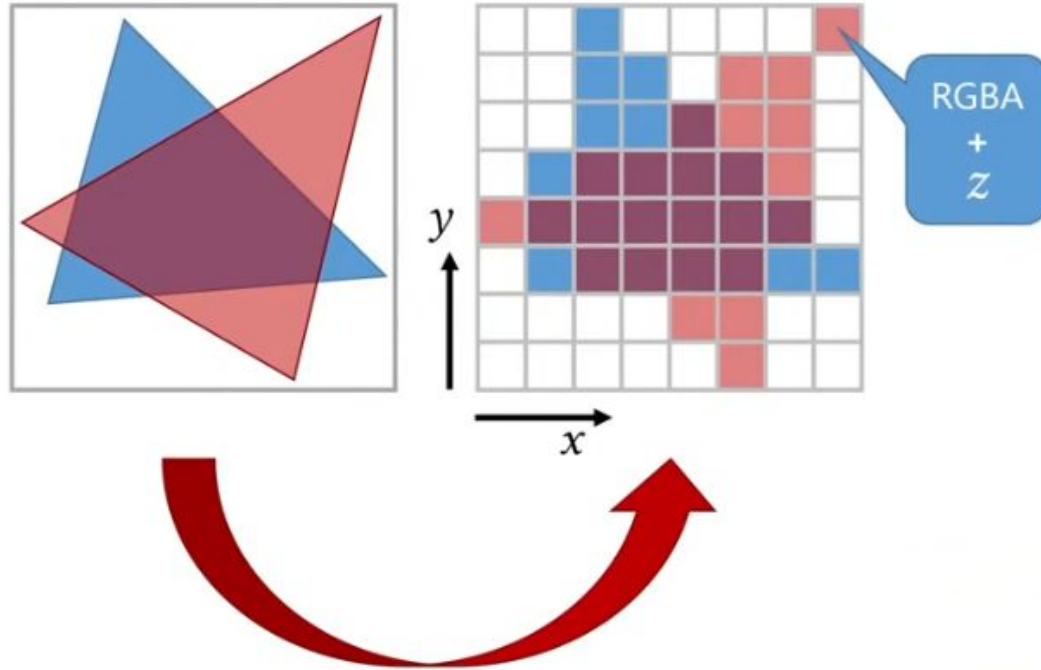
Transparency



Problem/s?

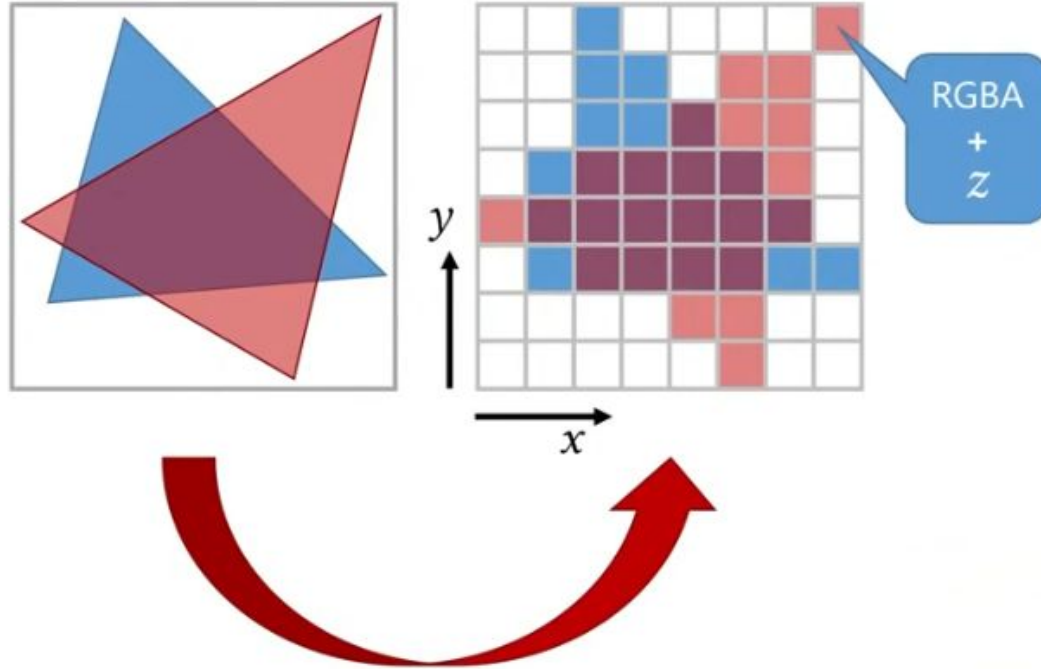
Transparency

Requires order



Problem/s?

Transparency
Requires order
Back to front



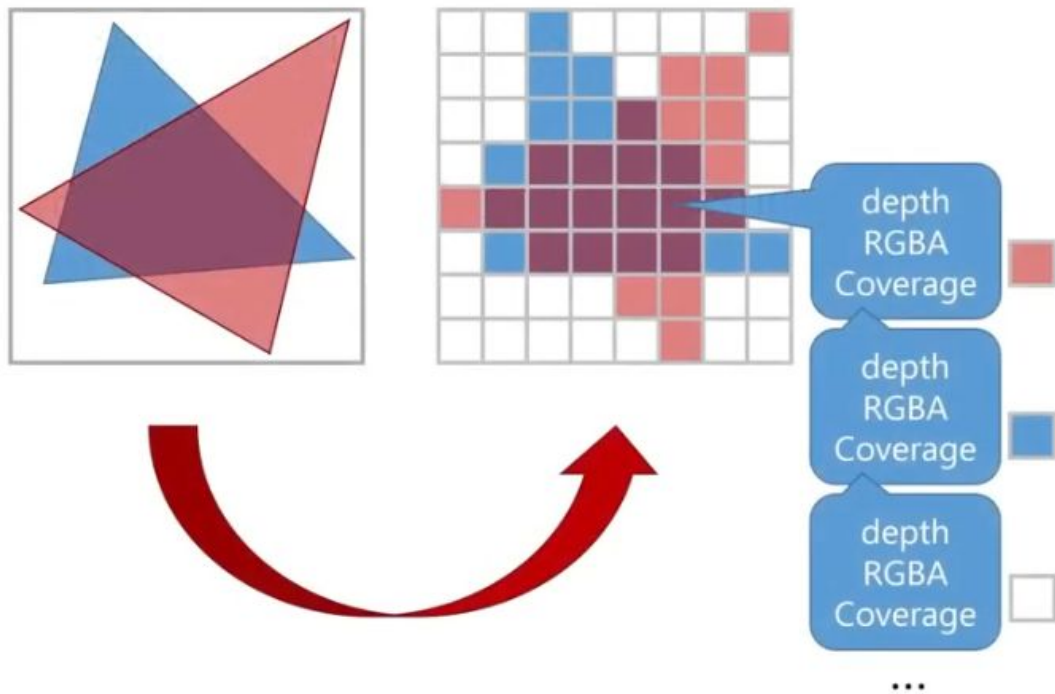
A-buffer

Handles intersection

Handles transparency irrespective of the ordering

But needs dynamic memory allocation (on GPU)

A-buffer



Render **E**verything **Y**ou **E**ver **S**aw



1986 Pixar Christmas Card by John Lasseter and Eben Ostby

Render **E**verything **Y**ou **E**ver **S**aw

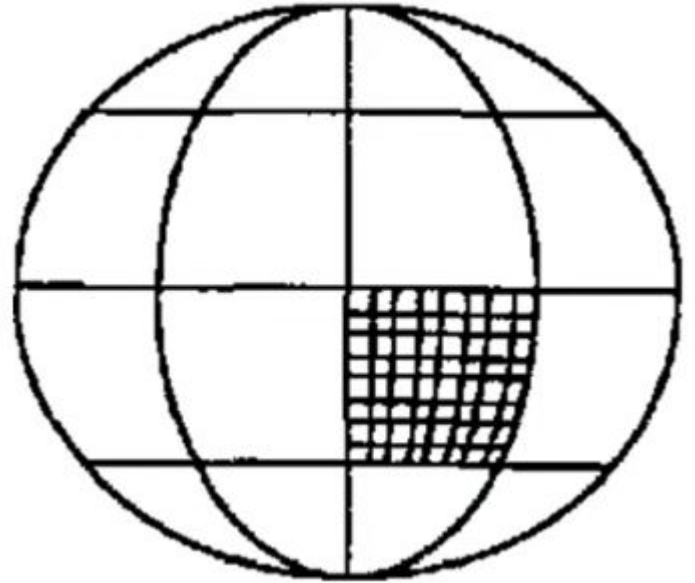
Use micropolygons

Data stored using bezier patches

Subdivide to small micropolygons.

Aggressive subdivision

Polygons become subpixel



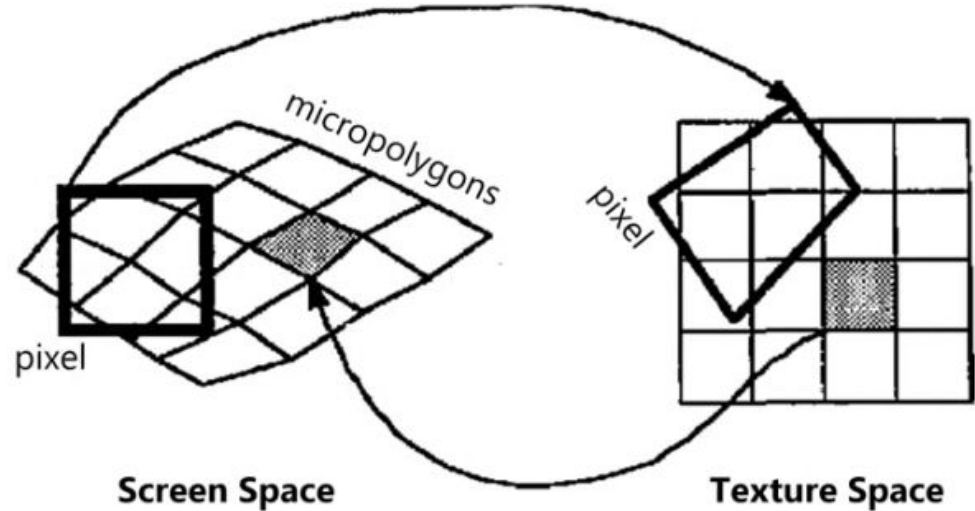
Render Everything You Ever Saw

Use micropolygons

Extremely costly

Still used by Pixar, Lucas films
etc

But in conjunction with Ray
tracing



Rasterization vs Ray tracing

For each primitive

Find the pixel which it maps to

For each pixel

Find all the primitive which contributes to that pixel

If objects are opaque, find the closest primitive